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Enhanced scaling crossover detection in long-range correlated time series

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ABSTRACT

Various time series, such as biological signals and stock prices, exhibit long-range correlations and a fractal nature characterized by the power-law scaling in the low-frequency range of the power spectrum. Instead of the power spectral analysis, scaling analysis methods such as Detrended Fluctuation Analysis (DFA) and Detrending Moving-Average Analysis (DMA) have been employed to estimate the scaling exponent. Scaling analysis results often uncover crossover phenomena, highlighting two distinct scaling regions—a short- and a long-range exponent. Estimating the two respective scaling exponents and the crossover point is crucial, as they can provide insight into different underlying mechanisms or dynamics operating at various scales. However, DFA and DMA with higher-order detrending tend to distort the time scales, and methods for accurately estimating the crossover have not been thoroughly investigated. This study addresses scale distortions in higher-order DFA and DMA by leveraging the relationship between the fluctuation function and the power spectrum of time series. We propose a method for crossover estimation using the Savitzky–Golay differentiation filter. Applying this method to numerical experiments with an autoregressive process exhibiting crossovers demonstrated that our proposed technique estimates crossovers more accurately than conventional segmented regression approaches used in DFA and DMA. We applied the proposed scaling crossover estimation method to the time series of the postural center of pressure (CoP), showcasing its practical applications in studying long-range correlations in empirical time series.

1. Introduction

Long-range correlations and the fractal nature of time series refer to persistent relationships between values separated by long time intervals. These can be characterized by fractal-like power-law scaling observed in the autocorrelation's long-time lag region and the power spectrum's low-frequency region. Unlike short-range correlations, which diminish quickly exponentially with a characteristic timescale, long-range correlations imply that distant points in the time series significantly depend on each other over a long period, making it impossible to define the characteristic timescale [1,2]. These correlations reveal critical insights into processes governing the system, enabling more accurate modeling and prediction. This phenomenon is evident in various domains, such as physiological signals, which may indicate complex biological processes [3–9]. It is also seen in financial markets, reflecting enduring market trends and investor behaviors [10–13]. To characterize such correlations observed in real-world time series with non-stationary trends, time-domain scaling analysis methods with detrending approaches, such as Detrended Fluctuation Analysis (DFA) [14–16], and Detrending Moving-Average Analysis (DMA) [17–21], have become popular to employ instead of conventional power spectrum analysis. In DFA and DMA, a scaling exponent that evaluates long-range correlations is estimated from the slope of the log–log plot, which regresses the root mean square of fluctuations against scale. However, estimates of these long-range correlations, quantified by a scaling exponent, often suffer due to the so-called "crossovers".

In the context of long-range correlations in time series analysis, a crossover refers to a shift in the scaling exponent across different timescales, indicating that the correlation behavior changes depending on the observed variation timescale of the time series. Usually, it arises by a change in the dynamical properties over different timescales [22–25]. For example, in physiological data, crossovers can reflect shifts from one type of regulatory mechanism to another, such as changes in heart rate variability or shifts in brain activity [25–31]. In stock prices, crossovers might indicate different market regimes or trading behaviors, such as transitioning from short-term speculative behavior to long-term investment strategies [32,33]. Understanding and accurately

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estimating these crossovers is crucial because they can reveal underlying mechanisms and dynamics that operate differently at various times [34,35]. This information can be valuable for developing better models, making predictions, or implementing interventions based on specific characteristics of the time series in different scaling regions.

Previously, the crossover point and two scaling exponents in the fluctuation function have been estimated using segmented regression [36]. This method of estimating regression models with visually perceptible breakpoints involves identifying and quantifying points at which the relationship between two variables – the logarithmic of scale and root mean square of fluctuations – changes significantly. The application of advanced algorithms, like those proposed by Muggeo [37], allows for detecting and estimating these breakpoints automatically; if this algorithm converges, significant breakpoints are believed to exist, but if the algorithm fails, it is possible that the breakpoint existed but was not detected. However, this simple regression method cannot accurately estimate two scaling exponents when the crossover changes gradually. For example, in physiological data such as heart rate variability, the crossover between different regulatory mechanisms might not be abrupt, leading to inaccuracies with segmented regression. Similarly, in financial time series, where market dynamics shift gradually rather than suddenly, broken-line regression may fail to accurately capture the scaling exponents' true nature. Therefore, a more effective approach is needed to detect crossovers in long-range correlations more accurately, even when the fluctuation function changes gradually.

Two critical steps precede accurate estimation of crossovers in long-range correlated time series. First, to reliably detect the scaling relationship in nonstationary time series with trends, it is crucial to distinguish deterministic trends from long-range correlations inherent in stochastic dynamics. Various detrending procedures have been introduced as best practices in scaling analysis to minimize the effects of trends [38–43]. Among these methods, DFA and DMA feature well-defined detrending processes and rigorous mathematical foundations [44–48]. Both methods include fluctuation functions with a clear relationship to the power spectrum of time series [47,48]. The original DMA corresponding to 0th-order DMA estimates the trend using a simple moving average [17,18]. A higher-order DMA has been proposed to generalize the 0th-order DMA, employing a higher-order moving average defined by a polynomial of degree m. This method is called the mth-order DMA (DMAm). Recent numerical and analytical studies have shown that higher-order DMA can remove higher-order polynomial trends embedded within the time series more effectively than DFA [44,49]. Because straightforward implementation of higher-order DMA is computationally expensive, which might prevent practical use of this method, a computationally efficient "fast" method has also been proposed [21].

The second, and perhaps more critical, challenge in estimating crossovers is the distortion of timescales caused by higher-order detrending. While higher-order detrending is necessary to remove deterministic trends, it invariably leads to significant scale distortion [46,47]. Indeed, the higher the order of detrending in DFA and DMA, the greater the distortion of scales. This issue arises because the relationship between time and frequency scales is disrupted, which should be given as a simple inverse relationship. Consequently, estimated crossovers might be shifted or obscured, leading to erroneous conclusions about the nature of correlations in the time series.

In this study, we leverage the relationship between the fluctuation function and power spectrum of time series to address scale distortions when estimating crossovers in long-range correlated time series using higher-order DFA or DMA. Specifically, we propose a method for crossover estimation using the Savitzky-Golay differentiation filter [50,51]. The Savitzky–Golay smoothing filter is an extended version of the simple moving-average filter using higher-order polynomials, and the higher-order filters can also give the differential coefficients of the smoothed signal. This work continues the tradition of refining multiscale analysis methods through improved filtering, demonstrating their validity for studying human physiology and other natural processes [52–54].

Using the proposed method, we conducted numerical experiments to confirm the accurate estimation of crossovers. We simulated time series using autoregressive fractionally integrated moving average (ARFIMA) processes, which are well-suited for modeling long-range correlations and crossover phenomena. (While ARFIMA was our chosen method, crossovers can also be effectively simulated using various other approaches [55].) ARFIMA models extend traditional ARIMA models by incorporating a fractional differencing parameter d, allowing for a more flexible representation of long-memory behavior. This parameter d can take non-integer values, capturing the persistence of correlations over longer periods. The general form of an ARFIMA process is given by ARFIMA(p, d, q), where p represents the order of the autoregressive part, d is the fractional differencing parameter, and q denotes the order of the moving average part. This model combines autoregressive (AR) and moving average (MA) components with fractional integration, making it highly effective in capturing nuanced statistical behaviors in real-world time series with long-range correlations and crossovers. Incorporating AR and ARFIMA processes ensured that these numerical experiments reflected some complexities and characteristics inherent in real-world time series [56,57].

To explore the advantages of crossover estimation using higher-order DFA and DMA with the Savitzky–Golay differentiation filter in the study of human physiology, we applied this method to the postural CoP time series in healthy young adults, older adults, and older adults with Parkinson's disease standing on a stable surface and a foam surface. Typically, postural CoP trajectory in upright stance exhibits long-range fluctuations and crossovers [58–62], indicating the transition between feedforward and feedback control of posture [63–65]. Therefore, we expected CoP fluctuations in healthy young adults to exhibit two distinct scaling exponents separated by a crossover. We hypothesized that standing on an unstable surface would enhance anticipatory control, causing a leftward shift of the crossover to longer scales and an increase in the long-range scaling exponent. Similarly, we expected that neurodegenerative conditions, such as Parkinson's disease, would decrease the long-range scaling exponent, reflecting reduced flexibility in bodily degrees of freedom in these patients. Combined with the above numerical experiments, success with this specific empirical case will establish the effectiveness of the proposed higher-order DFA and DMA with the Savitzky–Golay differentiation filter in accurately estimating crossovers. This will emphasize their utility in studying postural control across various task constraints and disease states.

The organization of this paper is as follows. Section 2 introduces procedures for scaling analysis. Section 3 addresses identifying and correcting scale distortions caused by detrending processes. In Section 4, we compare two crossover estimation methods: segmented regression and a combination of the Savitzky–Golay differentiation filter with kernel density estimation. Section 5.1 presents numerical experiments comparing DFA and DMA. Section 5.2 details the application of our crossover estimation method to postural CoP time series. A brief discussion follows this in Section 6. Finally, Section 7 summarizes the paper and offers concluding remarks. Full details of AR processes used in numerical simulations are provided in Appendix. We also refer readers to related work on this topic [34].

2. Scaling analysis

Before discussing crossover estimation, we introduce the fundamentals and methodology of scaling analysis.

2.1. Spectral analysis

In a stationary long-range correlated process, let us assume the autocorrelation follows

$$C(k) \sim k^{-\gamma}, \tag{1}$$

where $k \gg 1$ and $0 < \gamma < 1$. Our main task in the time series analysis of this process is to estimate the scaling exponent γ. According to the Wiener–Khinchin theorem, the power spectrum of this process is given by

$$\begin{aligned} S(f) &\sim f^{\gamma-1} \\ &\sim f^{-\beta}. \end{aligned} \tag{2}$$

Therefore, instead of autocorrelation estimation with a narrow estimable range of the scaling exponent, power spectrum analysis can be used to estimate the scaling exponent $\beta = 1 - \gamma$ as the slope of the double logarithmic plot of the power spectrum. Although the corresponding β range is from 0 to 1 due to $0 < \gamma < 1$ under the stationary assumption, the power spectrum can also estimate β for values outside this range. Accurate power spectrum estimation involves skilled techniques such as trend elimination, windowing, averaging over periodograms, and smoothing. Even with just windowing, many options, such as Hamming, Hanning, or Blackman windows, could confuse non-experts in spectral estimation. As an alternative to power spectrum estimation, non-experts increasingly use time-domain scaling analysis methods, such as DFA and DMA. In the following, we present analytically derived power spectra for stochastic process models but do not estimate power spectra using time series samples; the exact relation between DFA and DMA results and power spectrum has already been shown [47,48].

2.2. Detrended Fluctuation Analysis (DFA)

The procedures of DFA can be summarized in the following steps [24,25]:

1. *Integration of the observed time series*: Compute the cumulative sum of the time series. If the original time series is $x[i]$, then the integrated time series is $y[i] = \sum_{j=1}^{i}(x[j] - \bar{x})$, where $\bar{x}$ is the mean of the original time series, given by $\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x[i]$. This integration step can be skipped if the observed time series is diffusive, such as a sample of fractional Brownian motion. In that case, y is replaced by x in the following.
2. *Division of the time series into non-overlapping segments*: Divide the integrated time series $y[i]$ into non-overlapping segments of equal length s; the number of segments will be $l = \lfloor N/s \rfloor$, where N is the length of the integrated time series.
3. *Detrending of the time series*: For each segment, fit a polynomial of degree m (typically linear, but higher-order polynomials can be used for higher-order DFA) to the data in that segment. The method is called the mth-order DFA (DFAm), depending on the detrending order m. Let $y_{\mathrm{fit}}^{(l)}[i]$ be the fitted polynomial in the lth segment. Subtract the fitted polynomial from the integrated time series in each segment to obtain the detrended series:
$$y_{\mathrm{detrended}}^{(l)}[i] = y[i] - y_{\mathrm{fit}}^{(l)}[i]. \tag{3}$$
4. *Calculation of fluctuation function*: Compute the fluctuation function $F(s)$ as defined as the root-mean-square statistics of the detrended time series for each segment:
$$F(s) = \sqrt{\frac{1}{\lfloor N/s \rfloor}\sum_{l=1}^{\lfloor N/s \rfloor}\left\{\frac{1}{s}\sum_{k=(l-1)s+1}^{l\cdot s}\left(y_{\mathrm{detrended}}^{(l)}[k]\right)^2\right\}}. \tag{4}$$
5. *Repetition for multiple scales*: Repeat the detrending process for different scales s. Typically, s varies over various scales, often logarithmically equally spaced from $(m+1)$ to $\lfloor N/10 \rfloor$.
6. *Slope estimation*: Plot $\log_{10} F(s)$ versus $\log_{10} s$ and estimate the slope by least-squares fitting a line if a straight line is observed in the plot. The slope gives the estimated scaling exponent α, which characterizes the long-range correlation properties of the time series.
$$F(s) \sim s^{\alpha}. \tag{5}$$

When Eq. (1) holds, $F^2(s)$ is written as

$$\begin{aligned} F^2(s) &\sim s^{2-\gamma} \\ &\sim s^{2\alpha}. \end{aligned} \tag{6}$$

Furthermore, from Eqs. (2) and (6), the relationship between scaling exponents α and β is given by

$$\alpha = \frac{1+\beta}{2}. \tag{7}$$

2.3. Detrending Moving-Average Analysis (DMA)

The procedures of the DMA differ from those of the above-mentioned DFA in 2 and 3 but are otherwise identical [21]:

2. *Detrending of time series using the Savitzky–Golay smoothing filter*: In DMA analysis, an mth-order Savitzky–Golay smoothing filter [50,51] replaces the piecewise mth-order polynomial detrending used in DFA. An mth-order polynomial is fit to segments of scale s, using the center point of these segments as the filtering point. Specifically, it computes the filtered time series by evaluating the value of the least-squares polynomial at the central point within each segment. Given the equivalence between mth-order DFA and $(m+1)$th-order DMA results, m is set to an even value [44–46]. Accordingly, s must be an odd number. The Savitzky–Golay smoothing filter ensures that the time series retains the reproductive property of the normal distribution during the detrending process [50,51]. Detrending preserves its Gaussian nature if the original time series is a Gaussian process. However, DFA detrending can distort the Gaussian process, potentially transforming it into a non-Gaussian one, leading to less stable estimation results. Our previous study [66] demonstrated the superior detrending capabilities of the Savitzky–Golay smoothing filter compared to other conventional methods—it effectively removes higher-order polynomial trends from the original time series.
To apply the Savitzky–Golay smoothing filter, the smoothly varying component $\tilde{y}_s[i]$ is approximated by the Savitzky–Golay smoothing filter and removed from $y[i]$:
$$y_{\mathrm{detrended}}^{(s)}[i] = y[i] - \tilde{y}_s[i]. \tag{8}$$
3. *Calculation of fluctuation function*: Compute the root-mean-square fluctuation of the detrended time series for each segment:
$$F(s) = \sqrt{\frac{1}{N-s+1}\sum_{k=(s+1)/2}^{N-(s-1)/2}\left(y_{\mathrm{detrended}}^{(s)}[k]\right)^2}. \tag{9}$$

A direct implementation of the aforementioned DMA procedure is computationally intensive and impractical. For more efficient execution, the fast algorithm proposed by Tsujimoto et al. [66] offers a significantly better alternative. While the straightforward approach has a computation time proportional to N^2, the fast algorithm reduces this to just N.

3. Scale distortion and correction in scaling analysis

The location of the crossover corner indicates the characteristic timescale of the dynamics underlying the observed time series. To interpret this timescale accurately, accounting for and correcting the scale distortion introduced by the detrending operations in DFA and DMA is essential. This section outlines the process of correcting this scale distortion.

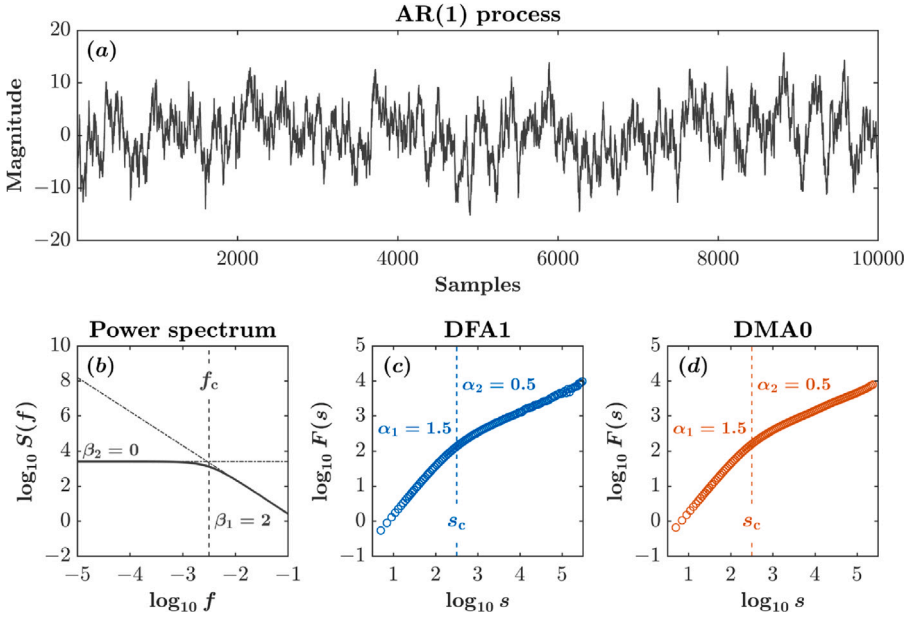


Fig. 1. Illustration of a correlated time series with crossovers. (*a*) Time series generated using an AR(1) process with $\log_{10} f_c = -2.5$ and $\log_{10} s_c = 2.5$ (length, $N = 100{,}001$ samples; only 10,000 samples are plotted in (*a*) for clarity; see Appendix). (*b*) Power spectral density of the time series with a flat white-noise plateau for $|f| \to 0$ and a $1/f^2$ behavior for the high-frequency range. (*c*) Fluctuation function of the time series yielded by DFA1. (*d*) Fluctuation function of the time series yielded by the equivalent DMA0. Dashed vertical lines in (*b–d*) indicate crossovers. β_1 and β_2 in the frequency domain, and α_1 and α_2 in the time domain, represent two scaling exponents separated by the crossovers.

Fig. 1 shows theoretical predictions of the power spectrum (***b***) and outcomes of applying DFA1 (***c***) and DMA0 (***d***) analysis to a sample time series (***a***) generated using the AR(1) process with crossover frequency $f_c = 10^{-2.5}$. Notice that f_c is at the boundary of two regions, a flat white-noise plateau for $|f| \to 0$ and $1/f^2$ behavior in the high-frequency range. For the lowest-order DFA and DMA, the inverse relationship, $s_c = 1/f_c$, holds, meaning $s_c \approx 1/10^{-2.5} = 10^{2.5}$. By definition, the AR(1) process has a Hurst exponent of 0.5.

In Fig. 1, *b* & *c*, we observe a crossover with two distinct regimes. The small-scale regime has $\alpha_1 = 1.5$. One can argue that this behavior can reflect finite-size effects [67], which can distort scaling exponents at short scales. However, we have analytically demonstrated that this effect depends on the order of DMA and the Hurst index, typically manifesting at scales smaller than 10 points [44]. Importantly, we established that the crossover induced by the finite-size effect arises from a methodological limitation rather than being an intrinsic property of the stochastic process itself. In contrast, the crossover phenomena investigated in the present work are fundamentally driven by the finite memory effect of the stochastic process. An AR(1) process, such as $x[i] = ax[i-1] + \epsilon[i] \quad (|a| < 1)$, exhibits asymptotically uncorrelated behavior characterized by a Hurst exponent of 0.5. However, when $a \to 1$, the process demonstrates Brownian motion-like behavior with a Hurst exponent of 1.5 within the scale range less than $\frac{2\pi\sqrt{a}}{1-a}$. Beyond this characteristic scale, the memory effect governed by a vanishes.

However, higher-order DFA and DMA have different detrending operations, resulting in distinct scale distortions. Fig. 2, *a* & *b* shows the application of higher-order DFA and DMA to the AR(1) process with $f_c = 10^{-1.5}$. Increasing the detrending order in both methods exacerbates scale distortion by disrupting the relationship, $s_c = 1/f_c$. This distortion arises mainly from the frequency response of the detrending operation. Consequently, estimated crossovers may be shifted or obscured, leading to incorrect conclusions about correlations in the time series. Ideally, this relationship should remain one-to-one, so this distortion must be corrected.

Below, we derive frequency responses of detrending operations of DFA and DMA to correct scale distortions. Consider the frequency response when the observed time series $x[i]$ is input and the integrated and detrended time series $y^{(s)}_{\text{detrend}}[i]$ is output in DFA and DMA algorithms. To do this, we refer to methods proposed in [47]; for DMA, frequency response can be computed using the same approach as for a linear system; for DFA, the linear system approach cannot be used because of nonlinear detrending, but a time domain approach to computing a single frequency response can be used to derive the pseudo-frequency response.

As shown in Nakata et al. [68], the detrended time series at scale s in DMA is reformulated as

$$y^{(s)}_{\text{detrended}}[i] = \sum_{k=1}^{s} w_s(k) x[i+k], \tag{10}$$

where $\{w_s(k)\}$ are given by

$$w_s(k) = \begin{cases} \Theta\left(\frac{s+1}{2} - k\right) - \sum_{j=0}^{m} b_{1,j+1} \sum_{r=k}^{s} (r - s' - 1)^j & 1 \le k \le s \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

In Eq. , $\Theta(x)$ is the unit step function, which equals 0 for $x < 0$ and 1 for $x \ge 0$, and $b_{i,j}(s)$ are elements of the inverse matrix of

$$B_m(s) = \sum_{j=-(s-1)/2}^{(s-1)/2} \begin{bmatrix} 1 & 0 & j^2 & \cdots & j^m \\ 0 & j^2 & 0 & \cdots & 0 \\ j^2 & 0 & j^4 & \cdots & j^{m+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ j^m & 0 & j^{m+2} & \cdots & j^{2m} \end{bmatrix}. \tag{12}$$

The $\{w_s(k)\}$ can be obtained by convolutioning the integral and detrending operations in DMA because both operations can be expressed in linear convolution forms [68]. In DMA0, we get

$$w_s(k) = \Theta\left(\frac{s+1}{2} - k\right) + \frac{k-s-1}{s}, \tag{13}$$

and in DMA2,

$$w_s(k) = \Theta\left(\frac{s+1}{2} - k\right) - \frac{(k-s-1)\left(10k^2 - 5ks - 20k - 2s^2 + 5s + 18\right)}{2s\,(s+2)\,(s-2)}. \tag{14}$$

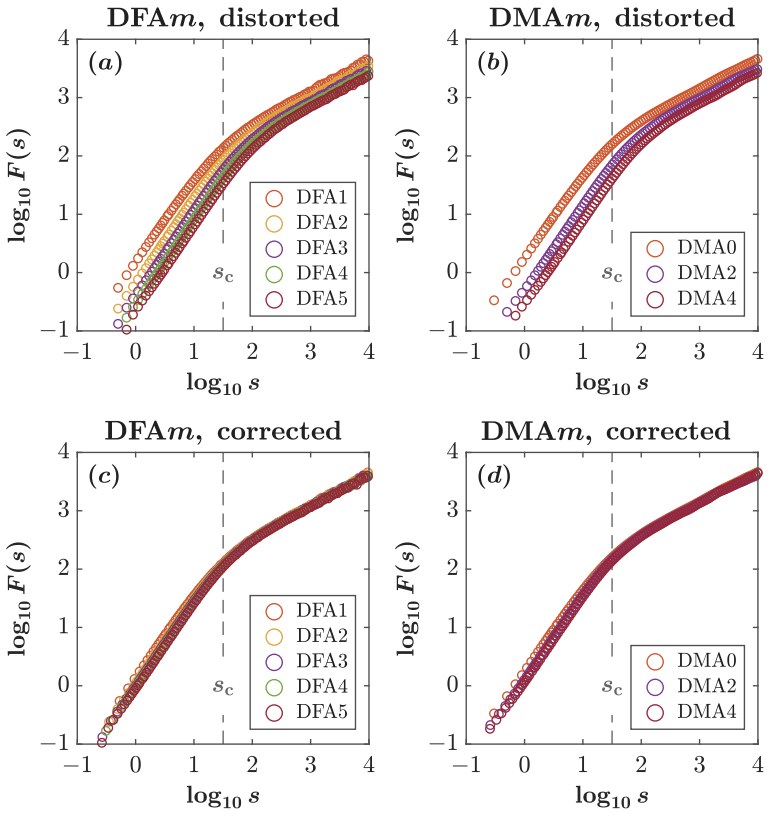


Fig. 2. Scale distortion and correction using DFAm and DMAm on time series generated using an AR(1) process with $\log_{10} f_c = -2.5$ and $\log_{10} s_c = 2.5$ (length, $N = 100{,}001$ samples; see Appendix). (a, b) Scale distortion in DFAm and DMAm before correction. (c, d) After correction by the distortion ratio r_d, scales align in DFAm and DMAm. Dashed lines in a–d indicate crossovers.

Using $\{w_s(k)\}$ and based on the convolution theorem, the frequency response can be calculated as

$$G_s(f) = \sum_{k=1}^{s} w_s(k) \exp(-i2\pi f k), \tag{15}$$

where i is the imaginary unit. If the power spectrum of a discrete stochastic process is given by $S_x(f)$, the corresponding fluctuation function $F^2(s)$ can be calculated analytically by

$$\begin{aligned} F^2(s) &= \frac{1}{N} \sum_{k=0}^{N-1} S_x\left(\frac{k}{N}\right) \left|G_s\left(\frac{k}{N}\right)\right|^2 \\ &= \int_{-1/2}^{1/2} S_x(f) \left|G_s(f)\right|^2 \, df \quad (N \to \infty). \end{aligned} \tag{16}$$

This equation clearly shows that $\left|G_s(f)\right|^2$ relates the frequency f to the timescale s. Although Eq. (16) also means that $F^2(s)$ is equivalent to $S_x(f)$, since $\left|G_s(f)\right|^2$ smoothens the noisy power spectrum estimate (periodogram), $F^2(s)$ shows a smoother trace than $S_x(f)$ in practical analysis.

The detrended time series in DFA cannot be written as in Eq. (10) because the detrending operation using piecewise fitting cannot be expressed in linear convolution form. Therefore, unlike the above approach, we analytically calculate the single-frequency response function $\overline{\Phi}^2(s, f, A)$ defined as the mean-square deviation from the trend when a single-frequency component

$$x^{(f)}[i] = A \cos(2\pi f i + \theta), \tag{17}$$

with frequency f and amplitude A is directly analyzed by DFA [47]. In this case, the integrated series of $x^{(f)}[i]$ is given by

$$y^{(f)}[i] = \frac{A \sin(2\pi f i - f\pi + \theta)}{2 \sin(\pi f)}, \tag{18}$$

where the constant term is removed to simplify the calculations. To fit mth-order polynomial $y^{(l)}_{\mathrm{fit}}[i] = a_0 + a_1 i + \cdots + a_m i^m$ in the segment (window) over $[0, s-1]$, we minimize

$$I\left(\{a_0, a_1, \ldots, a_m\}\right) = \sum_{i=0}^{n-1} \left\{ y^{(f)}[i] - y^{(l)}_{\mathrm{fit}}[i] \right\}^2, \tag{19}$$

and then, using the obtained coefficients $\{a_0, a_1, \ldots, a_m\}$, the mean-square deviation is calculated by

$$\Phi^2(s, f, A, \theta) = \frac{1}{s} \sum_{i=0}^{s-1} \left\{ y^{(f)}[i] - \left(a_0 + a_1 i + \cdots + a_m i^m\right) \right\}^2. \tag{20}$$

Finally, by averaging phase θ in this equation over $[0, 2\pi]$, we obtain the single-frequency response function $\overline{\Phi}^2(s, f, A)$. For instance, $\overline{\Phi}^2(s, f, A)$ of DFA1 is given by

$$\begin{aligned} \Phi^2(s, f, A, \theta) = \frac{A^2}{64 s^2 \left(s^2 - 1\right) \sin^6(\pi f)} \Big\{ & \left(s^2 - 1\right) s^2 \cos(4\pi f) \\ & + 2s^2 \cos(2\pi f (s-1)) - 4s^2 \cos(2\pi f s) \\ & + 2s^2 \cos(2\pi f (s+1)) - 4\left(s^4 - 3s^2 + 2\right) \cos(2\pi f) \\ & + 6s \cos(2\pi f (s-1)) - 6s \cos(2\pi f (s+1)) \\ & + 4 \cos(2\pi f (s-1)) + 4 \cos(2\pi f s) \\ & + 4 \cos(2\pi f (s+1)) + 3s^4 - 11 s^2 - 4 \Big\}. \end{aligned} \tag{21}$$

If the power spectrum of a discrete stochastic process is given by $S_x(f)$, then the corresponding fluctuation function $F^2(s)$ can be calculated analytically as

$$\begin{aligned} F^2(s) &= \frac{1}{N} \sum_{k=0}^{N-1} \overline{\Phi}^2\left(s, f = \frac{k}{N}, A = \sqrt{S_x\left(\frac{k}{N}\right)/N}\right) \\ &= \int_{-1/2}^{1/2} S_x(f) \, \overline{\Phi}^2(s, f, A = 1) \, df \quad (N \to \infty). \end{aligned} \tag{22}$$

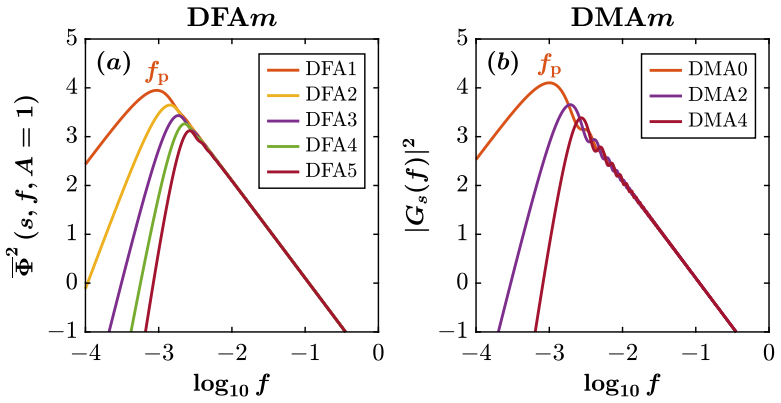


Fig. 3. The relationship between power spectrum $S(f)$ and fluctuation function $F(s)$ can be analytically determined to obtain the frequency response function with peak frequency f_p. (*a*) $\overline{\Phi}^2(s, f, A = 1)$ of DFA*m*. (*b*) $|G_s(f)|^2$ of DMA*m*. Scale $s = 1{,}000$.

Table 1
Distortion ratio r_d for *m*th-order DFA and DMA obtained by numerical experiments.

r_d	0th	1st	2nd	3rd	4th	5th
DFA	N/A	0.94	1.43	1.86	2.26	2.65
DMA	1	N/A	1.93	N/A	2.74	N/A

This equation is the same form as Eq. (16). Therefore, in the case of DFA, we evaluate $\overline{\Phi}^2(s, f, A = 1)$ instead of $|G_s(f)|^2$.

To correct the scale distortion, we define the distortion ratio as

$$r_d = \frac{s}{1/f_p}, \tag{23}$$

where f_p is estimated as the peak frequency of $|G_s(f)|^2$ in Eq. (16) for DMA and $\overline{\Phi}^2(s, f, A = 1)$ in Eq. (22) for DFA (Fig. 3). Table 1 shows numerically estimated r_d. Both even-order *m*th and odd-order $(m+1)$th DMA represent the same detrending features, resulting in the same distortion ratio r_d. In the time domain, scale s is corrected as

$$\tilde{s} = \frac{s}{r_d}, \tag{24}$$

Fig. 2, c & d show the results after correcting for scale distortion.

4. Estimating crossovers in fluctuation functions

After a brief overview of conventional segmented regression, we introduce our proposed method, which incorporates the Savitzky–Golay differentiation filter and kernel density estimation for enhanced crossover detection.

4.1. Estimating crossovers using segmented regression

Traditionally, segmented regression has been used to estimate crossovers in fluctuation functions [37], wherein the segmented regression line is defined by

$$y[i] = \begin{cases} a_1 x[i] + b_1 & 1 \le i \le j \\ a_2 x[i] + b_2 & (j+1) \le i \le N, \end{cases} \tag{25}$$

where N is the number of points defining the line. A minimization process then follows:

$$e[j] = \sum_{i=1}^{j} \left\{ y[i] - a_1 x[i] - b_1 \right\}^2 + \sum_{i=j+1}^{N} \left\{ y[i] - a_2 x[i] - b_2 \right\}^2, \tag{26}$$

such that the crossover point is

$$x_c = \underset{x[i]}{\operatorname{argmin}} \left\{ e[j] \right\}. \tag{27}$$

When estimating crossovers, $x = \log_{10} s$ and $y = \log_{10} F(s)$, with estimated values including $\hat{a}_1 = a_1$, $\hat{a}_2 = a_2$, and $\log_{10} \hat{s}_c = x_c$ (see Fig. 4). To mitigate the sensitivity of segmented regression to gradual slope changes, we propose a crossover estimation method that combines the Savitzky–Golay differentiation filter with kernel density estimation. We demonstrate the advantages of this approach by comparing its estimation results with those of conventional segmented regression.

4.2. Estimating crossovers using a combination of the Savitzky–Golay differentiation filter and kernel density estimation

Once we have corrected for scale distortion, the next step is to detect crossovers in the fluctuation function. We propose a novel method for estimating crossovers using a combination of the Savitzky–Golay differentiation filter [50,51] and kernel density estimation [69,70]. When a crossover occurs, it manifests as two regions with distinct slopes (scaling exponents), with the slope changing at the crossover point. This method derives the local slope profile from the 1st derivative coefficient obtained from Savitzky–Golay smoothing filtering (Fig. 5, *a* & *d*). We then estimate the two slopes ($\hat{a}_1$ and $\hat{a}_2$) and the crossover scale ($\hat{s}_c$) using kernel density estimation.

First, we obtain the 1st derivative coefficient at each scale s using the Savitzky–Golay differentiation filter (Fig. 5, *b* & *e*). In the Savitzky–Golay differentiation filter, polynomials fitted by the Savitzky–Golay smoothing filter are differentiated, and the center values of the differentiated polynomials provide the 1st derivative coefficients. The filter's window width is $w = 3, 7, 11, \ldots, \lfloor N/2 \rfloor$. To apply the Savitzky–Golay filter, we resample $\log_{10} F(s)$ of DFA and DMA as evenly-spaced values for $\log_{10} s$. Next, we estimate the distribution of the 1st derivative coefficient using kernel density estimation and identify the peak of this distribution as the scaling exponent (Fig. 5, *c* & *f*). A triangular kernel density function is employed due to its simplicity and efficiency. The triangular kernel is computationally less intensive than other kernel functions like Gaussian or Epanechnikov and has finite support, which reduces boundary bias and provides a smooth distribution estimate. Finally, we determine the scale of the 1st derivative coefficient where a local minimum occurs in the distribution and consider the peak's position as the crossover scale. Estimating the distribution of the second derivative coefficient can further refine the estimation process.

A critical observation in our analysis is that local slope computation using the Savitzky–Golay differentiation filter shows greater variation in local slope for longer scales in DFA1 compared to Detrending Moving-Average Analysis DMA0. This is evident from the window width-dependent kernel density distribution observed in DFA, where the density distribution varies significantly with changes in window size. In contrast, kernel density estimation for DMA remains consistent

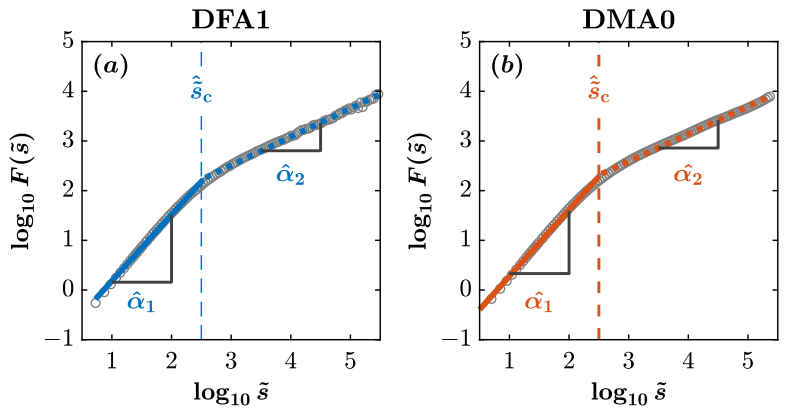


Fig. 4. Illustration of crossover estimation using segmented regression [36]. (*a*) Results for DFA1. (*b*) Results for DMA0. Dashed vertical lines indicate crossovers (i.e., $\log_{10} \hat{\tilde{s}}_c$). Solid lines indicate the best fit for $\log_{10} \tilde{s} \leq \log_{10} \hat{\tilde{s}}_c$, and dashed lines indicate the best fit for $\log_{10} \tilde{s} > \log_{10} \hat{\tilde{s}}_c$.

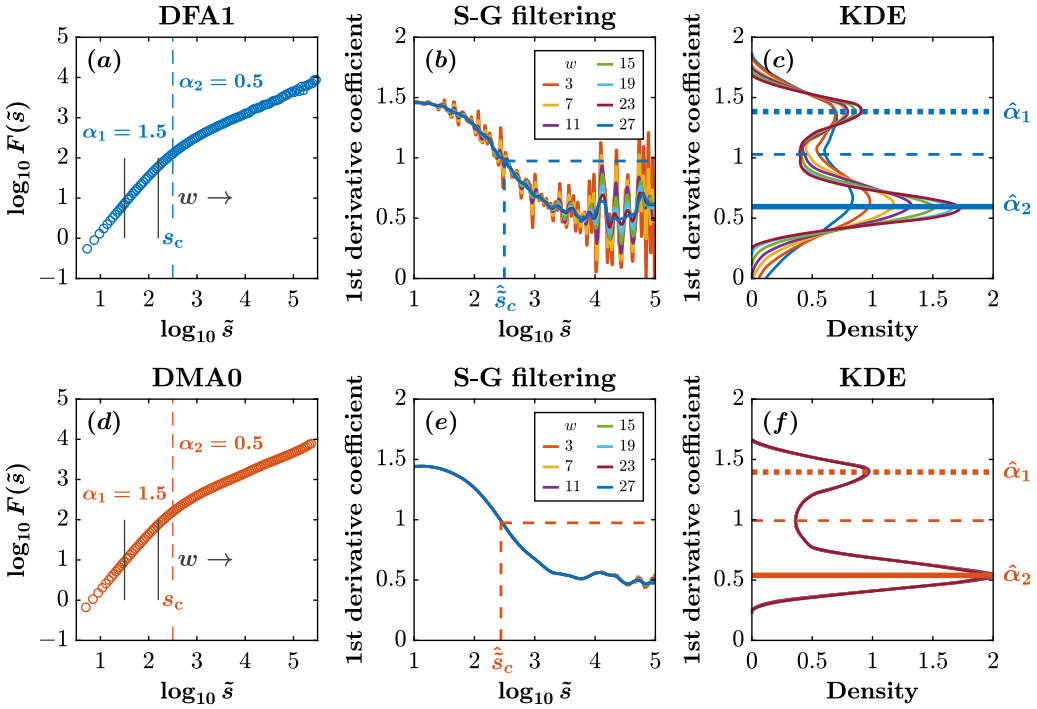


Fig. 5. Illustration of crossover estimation using a combination of Savitzky–Golay (S–G) differentiation filter and kernel density estimation (KDE). (*a*) Fluctuation function of an AR(1) process with $\log_{10} s_c = 2.5$ (length, $N = 100{,}001$ samples) obtained using DFA1. (*b*) 1st derivative coefficient of the S–G filtering obtained while changing window width $w = 3, 7, 11, \ldots, \lfloor N_f/8 \rfloor$, where N_f is the number of points in the fluctuation function. (*c*) Density distribution of the 1st derivative coefficient of the S–G filter used to estimate crossover scale, $\hat{\tilde{s}}_c$, as the scale, $\tilde{s}$, corresponding to the 1st derivative coefficient at the trough of the density distribution and to estimate two scaling exponents, $\hat{\alpha}_1$ and $\hat{\alpha}_2$, as peaks of the density distribution. (*d–f*) Same process for estimating crossover scale and two scaling exponents is replicated for the fluctuation function of the same AR(1) process obtained using DMA0. Dashed vertical lines in (*a*, *b*, *d*, & *e*) indicate crossovers. Dotted horizontal lines in (*c* & *f*) indicate $\hat{\alpha}_1$. Solid horizontal lines in (*c* & *f*) indicate $\hat{\alpha}_2$.

across different window sizes, suggesting more stable slope estimates. This consistency implies that DMA is more effective in estimating crossovers and distinct scaling exponents within the same fluctuation function in long-range correlated time series. Reduced sensitivity to window size in DMA indicates that it provides a more robust and reliable method for analyzing scaling properties in complex time series, making it superior to DFA in capturing nuanced variations in scaling behavior. This distinction between the two processes requires systematic investigation to understand their full implications for time series analysis. We used this combination of the Savitzky–Golay differentiation filter and kernel density estimation to estimate crossovers in numerically simulated time series and empirically obtain postural CoP trajectories exemplifying postural control in healthy young adults, healthy older adults, and older adults with Parkinson's disease.

5. Results

5.1. Numerical experiments

We generated 100 iterations of $N = 1{,}000{,}001$ sample-long time series with different random seeds using the AR(1), ARFIMA$(1, d, 0)$, and ARFIMA$(2, d, 0)$ processes. See Appendix for details of each process. Fig. 6 illustrates an example time series for each process, its corresponding power spectrum, and the fluctuation function obtained using DMA4. We set the parameter d to 0.4, making each time series correlated for a long time. We used the scaling region $1.0 \leq s \leq 5.0$ for scaling analysis. We then estimated the crossover scale ($\hat{s}_c$) and scaling exponents ($\hat{\alpha}_1$ and $\hat{\alpha}_2$) for each time series using both segmented and Savitzky–Golay methods. We calculated the mean and standard deviation of these estimates. First, we performed these steps for the

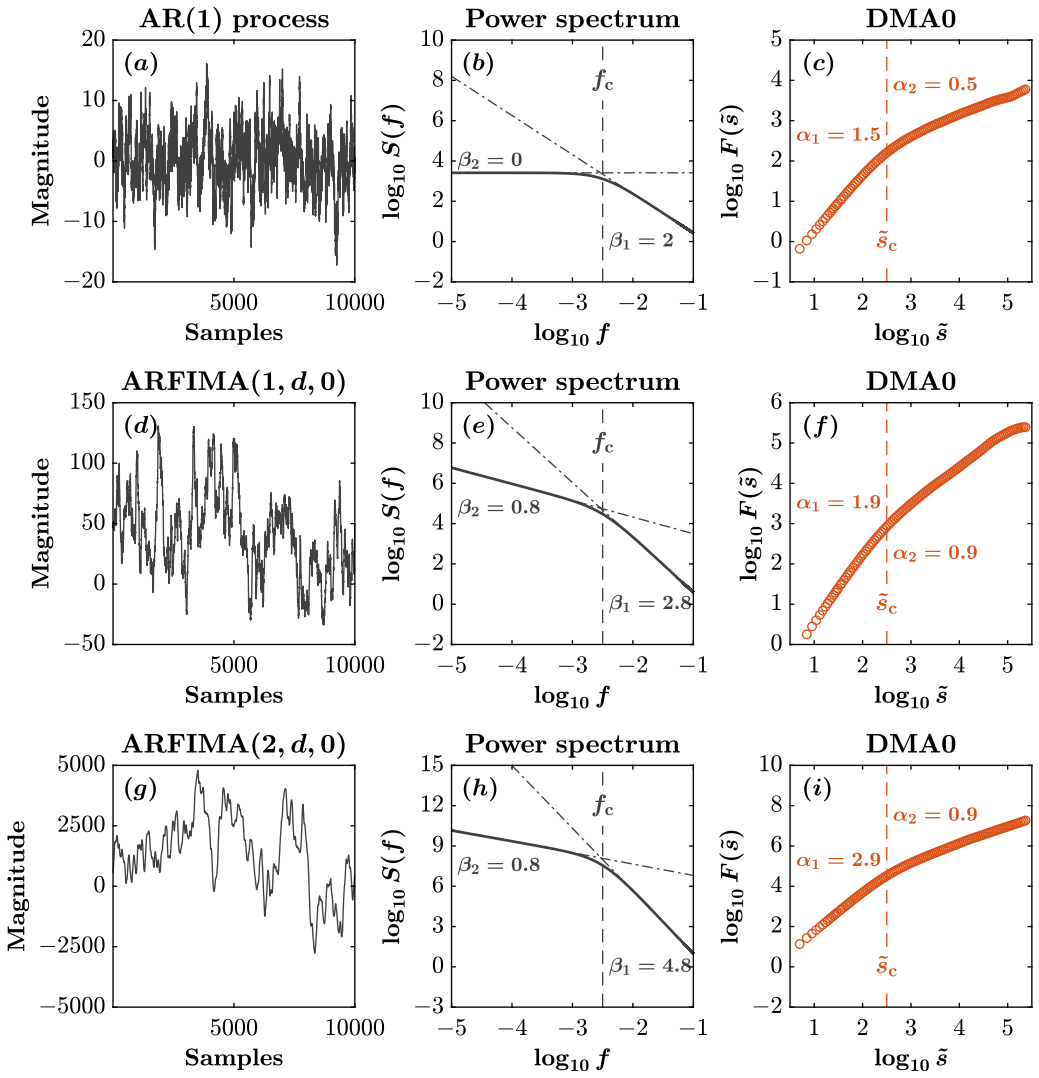


Fig. 6. Sample time series used for numerical experiments, along with their power spectral density and fluctuation functions obtained using DMA0 (length, $N = 100{,}001$ samples; see Appendix). (*a–c*) An AR(1) process with $\log_{10} f_c = -2.5$ and $\log_{10} s_c = 2.5$. (*d–f*) An ARFIMA$(1, d, 0)$ process with $\log_{10} f_c = -2.5$, $\log_{10} s_c = 2.5$, and $d = 0.4$. (*g–h*) An ARFIMA$(2, d, 0)$ process with $\log_{10} f_c = -2.5$, $\log_{10} s_c = 2.5$, and $d = 0.4$.

AR(1) process with $\log_{10} s_c = 2.5$, $\alpha_1 = 1.5$, and $\alpha_2 = 0.5$. Fig. 7 describes these results, revealing that higher-order detrending better estimates the crossover scales ($\hat{s}_c$) and scaling exponents ($\hat{\alpha}_1$ and $\hat{\alpha}_2$). Accordingly, we used DMA4 to make further comparisons across all three processes.

We compared segmented regression and Savitzky–Golay differentiation filter methods for crossover estimation using DMA4. In each AR process, we varied the crossover scale s_c. Fig. 8, *a–c* present crossover estimation results across 100 simulations of each time series type for each value of s_c. Segmented regression showed decreasing accuracy in crossover estimation as the crossover scale deviated from the center of the log–log plot, with a more significant drop for shorter scales. In contrast, using the Savitzky–Golay differentiation filter, crossover estimation provided consistently accurate estimates independent of the crossover scale. Fig. 8, *d–i* demonstrate that both methods estimate $\hat{\alpha}_1$ and $\hat{\alpha}_2$ more accurately with a sufficiently wide scaling region. However, the Savitzky–Golay differentiation filter yields more precise scaling exponent estimates than segmented regression. Despite this, estimation accuracy declines in the ARFIMA$(2, d, 0)$ process across various scenarios.

As illustrated in Fig. 8, enhanced accuracy in estimating crossovers using higher-order DFA and DMA methods is tied to the inherent challenges in estimating scaling exponents with these techniques [46]. We recommend scale-corrected higher-order DFA and DMA for effective crossover estimation with sufficient trend rejection. The numerical experiments establish that the Savitzky–Golay differentiation filter offers superior estimates of scaling exponents compared to segmented regression, which is more sensitive to gradual changes in slope between α_1 and α_2. Unlike segmented regression, the Savitzky–Golay differentiation filter does not depend on regions with gradual changes and without distinct peaks in the distribution. Consequently, the Savitzky–Golay differentiation filter provides more accurate estimates of crossovers by detecting slope changes as smaller density variations in the distribution. In the case of ARFIMA$(2, d, 0)$, the pronounced change in slope highlights the greater disparity in estimation accuracy between segmented regression and the Savitzky–Golay differentiation filter for scaling exponents $\hat{\alpha}_1$ and $\hat{\alpha}_2$ compared to AR(1) and ARFIMA$(1, d, 0)$. These results indicate that the Savitzky–Golay differentiation filter method is well-suited for estimating crossovers. However, since this numerical experiment used ideal crossover scenarios, the next section will focus on applying this method to real data.

5.2. Application to postural control

Even in the so-called "quiet" stances, human posture exhibits intermittent fluctuations—sporadic, irregular bursts of corrective actions

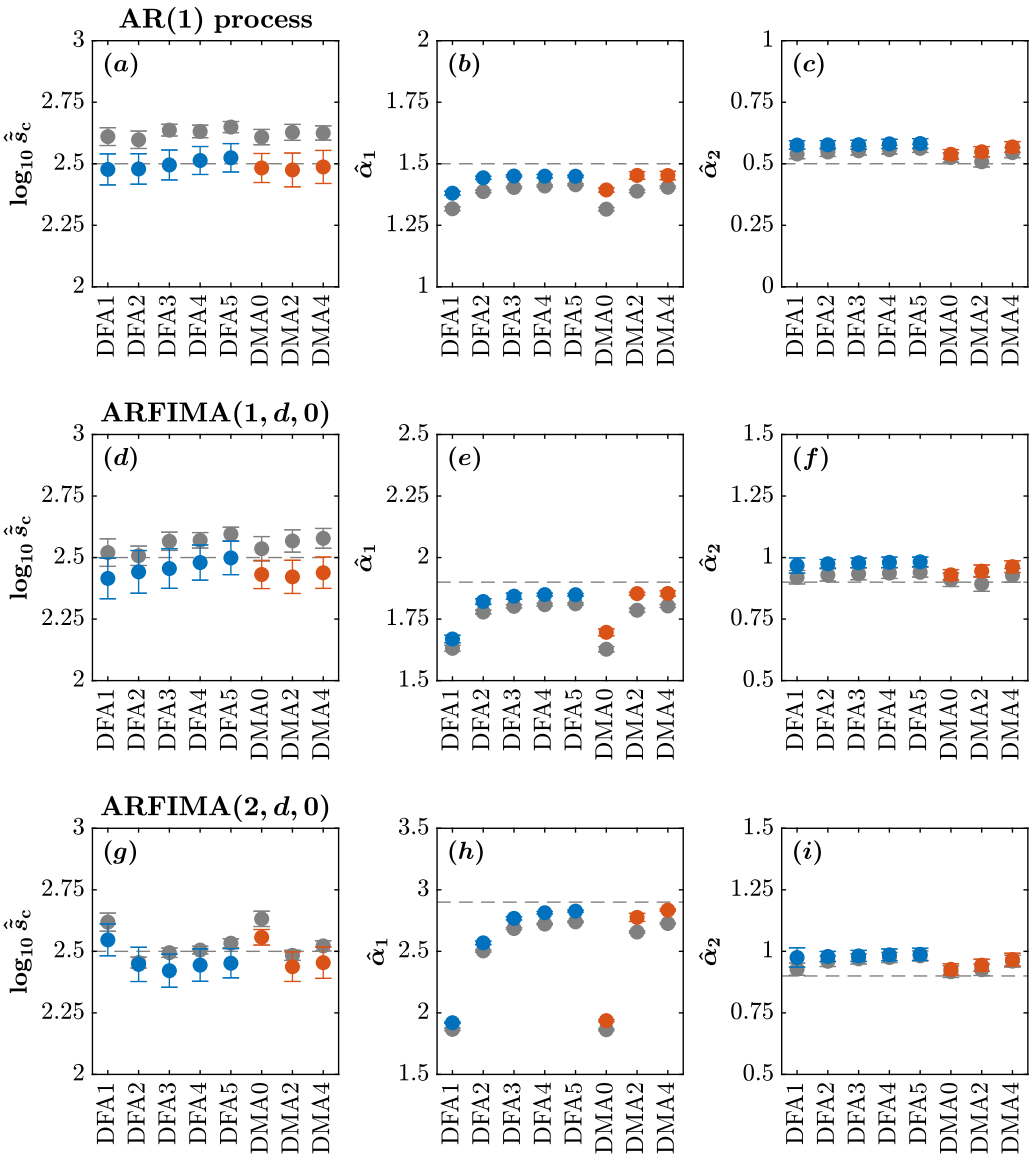


Fig. 7. Comparison of estimated crossover scales, $\hat{s}_c$, and two scaling exponents, $\hat{\alpha}_1$ and $\hat{\alpha}_2$, obtained using DFAm and DMAm with segmented regression and the combination of Savitzky–Golay differentiation filter and kernel density estimation. (*a*–*c*) Estimations for AR(1) processes with $\log_{10} f_c = -2.5$ and $\log_{10} s_c = 2.5$. (*d*–*f*) Estimations for ARFIMA$(1, d, 0)$ processes with $\log_{10} f_c = -2.5$, $\log_{10} s_c = 2.5$, and $d = 0.4$. (*g*–*i*) Estimations for ARFIMA$(2, d, 0)$ processes with $\log_{10} f_c = -2.5$, $\log_{10} s_c = 2.5$, and $d = 0.4$. Dashed horizontal lines in (*a*, *d*, & *g*) indicate theoretical $\log_{10} s_c$. Dashed vertical lines in (*b*, *e*, & *h*) indicate theoretical α_1, and dashed vertical lines in (*c*, *f*, & *i*) indicate theoretical α_2. Solid gray and blue/red circles indicate estimates obtained using segmented regression and the combination of Savitzky–Golay differentiation filter and kernel density estimation, respectively. Error bars indicate $\pm 1SD$ across 100 simulations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

interspersed with periods of relative stability that characterize the maintenance of an upright stance. In maintaining balance, the body's CoP exhibits fluctuations that are not continuous but marked by sudden, transient adjustments necessary to counteract the destabilizing effects of internal and external perturbations. Irregularity and unpredictability of these bursts suggest a sophisticated interplay between sensory feedback, neural processing, and motor output, ensuring balance is maintained through a dynamic and responsive control strategy [71–79]. During stable periods, CoP shows minimal movement, indicating effective control and equilibrium. However, when a perturbation occurs, the system responds with a burst of activity to restore balance. This pattern of intermittency is crucial for adaptive and flexible postural control, enabling the body to manage and respond to varying conditions and demands efficiently [80].

Long-range correlations and multiple scaling regions are well-documented phenomena in postural CoP dynamics [58–62]. Despite their counterintuitive nature associated with intermittency, these correlations are crucial for maintaining an upright stance. However, there is an ongoing debate about how this scaling relates to task context. One perspective suggests that distinct scaling regions arise from discrete switching between different control modes [63,65], where constraints on sway engagement or disengagement play a mediating role [81–83]. According to this viewpoint, constraints are disengaged when there is minimal sway and pose no immediate threat to stability, resulting in increased variability in sway. This randomness implies a stochastic sequence of constraint engagement and disengagement, leading to short-lag negative autocorrelation amidst uncorrelated variability. An alternative interpretation of crossovers in postural control focuses on feedback-based corrections. Postural control can exhibit two distinct behaviors: "trembling" around equilibrium points and "rambling", where shifts occur between different equilibrium points [84,85]. Rambling involves the integration of diverse information and necessitates

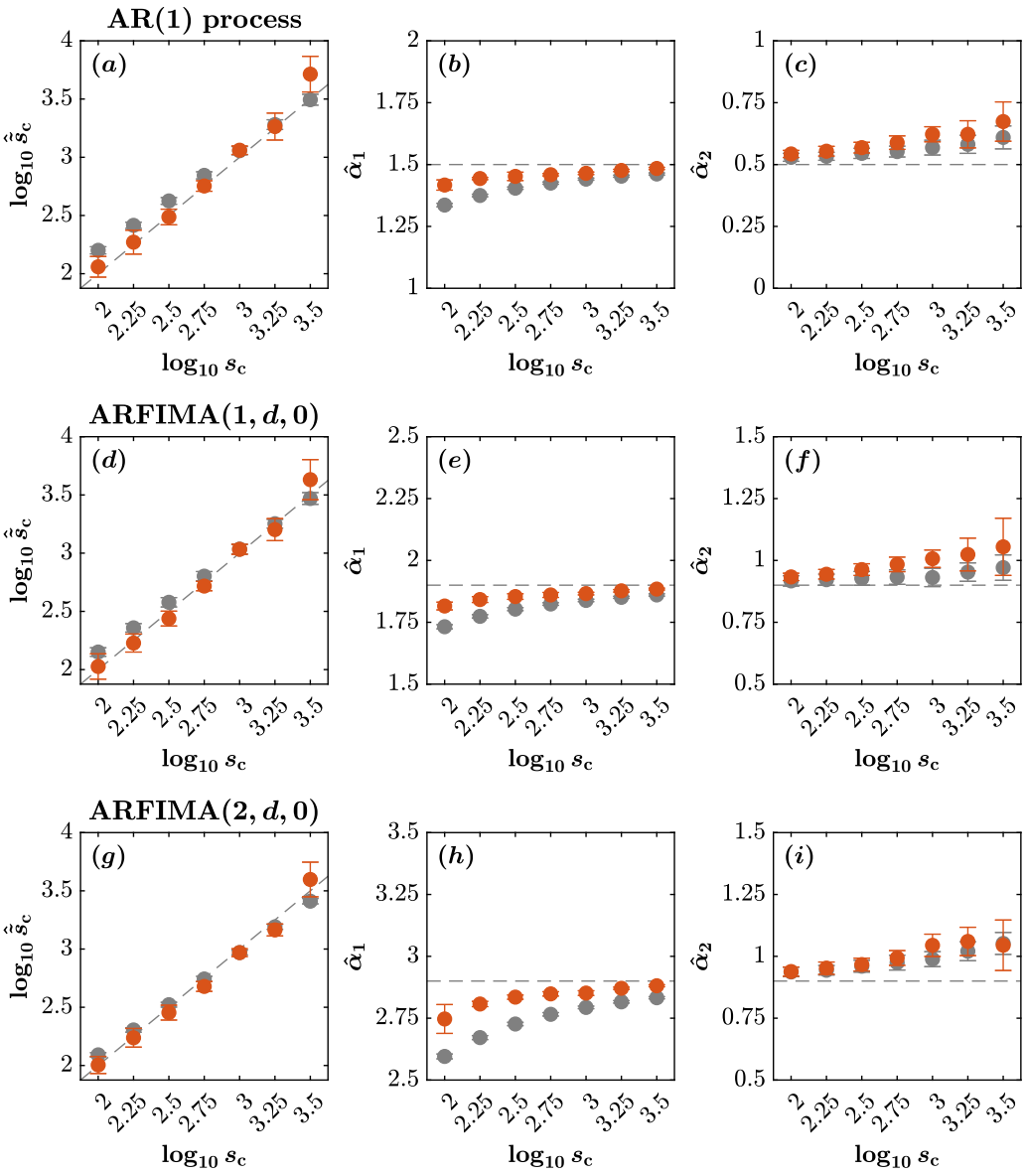


Fig. 8. Comparison of estimated crossover scales, $\hat{\tilde{s}}_c$, and two scaling exponents, $\hat{\alpha}_1$ and $\hat{\alpha}_2$, obtained using DMA4 with segmented regression and a combination of Savitzky–Golay differentiation filter and kernel density estimation. (*a*–*c*) Estimations for AR(1) processes with $\log_{10} f_c \in \{-3.5, -3.25, -3, \ldots, -2\}$ and $\log_{10} s_c \in \{2, 2.25, 2.5, \ldots, 3.5\}$. (*d*–*f*) Estimations for ARFIMA$(1, d, 0)$ processes with $\log_{10} f_c \in \{-3.5, -3.25, -3, \ldots, -2\}$, $\log_{10} s_c \in \{2, 2.25, 2.5, \ldots, 3.5\}$, and $d = 0.4$. (*g*–*i*) Estimations for ARFIMA$(2, d, 0)$ processes with $\log_{10} f_c \in \{-3.5, -3.25, -3, \ldots, -2\}$, $\log_{10} s_c \in \{2, 2.25, 2.5, \ldots, 3.5\}$, and $d = 0.4$. Dotted horizontal lines in (*a*, *d*, & *g*) indicate theoretical $\log_{10} s_c$. Dashed vertical lines in (*b*, *e*, & *h*) indicate theoretical α_1, and dashed vertical lines in (*c*, *f*, & *i*) indicate theoretical α_2. Solid gray and red circles indicate estimates obtained using segmented regression and the combination of the Savitzky–Golay differentiation filter and kernel density estimation, respectively. Error bars indicate $\pm 1SD$ across 100 simulations.

active mechanical processes [86–88], passive reflex mechanisms [89, 90], and anticipatory [91–93], feedback [94,95], and exploratory responses [96,97]. Coordinating these information sources across different spatial and temporal scales leads to observed crossovers in the scaling of CoP fluctuations.

To illustrate the application of the proposed method in real-life empirical measurements, we analyzed the fluctuation functions of two CoP trajectories from a healthy young adult standing on rigid and foam surfaces with eyes closed [98]. We focused on the first 30 seconds of the 60-second CoP trajectory recorded at 100 Hz, given the common use of 30-second assessments in the clinical domain. Fig. 9, *a* & *b* shows the CoP trajectory along the participant's anteroposterior axis while standing on rigid and foam surfaces, respectively, with eyes closed. The CoP trajectory for the rigid surface exhibited a clear crossover, $\log_{10} \tilde{s}$, that was visually perceptible (Fig. 9*c*). The 1st derivative coefficient of the fluctuation function also indicated a region with a drastic change in $\log_{10} F(\tilde{s})$ (Fig. 9*d*), and kernel density estimation confirmed a local minimum in this region, accurately identifying the crossover scale at $\log_{10} \hat{\tilde{s}}_c = 1.93$ (Fig. 9*e*). The two peaks of the density distribution reflected the two scaling exponents, $\hat{\alpha}_1 = 2.67$ and $\hat{\alpha}_2 = 0.55$, encoding—the speed of CoP at shorter and longer scales. This crossover at ~ 850 milliseconds indicates a shift in the nervous system's postural control strategies. Balance is maintained at scales shorter than 850 milliseconds through rapid, reflexive adjustments, resulting in faster CoP displacements. Conversely, longer scales involve more long-term control processes, such as anticipatory and strategic adjustments by the nervous system, leading to slower CoP displacements.

In contrast, the CoP trajectory for the foam surface exhibited a relatively gradual fluctuation function, making it difficult to visually identify the crossover(s) (Fig. 9*f*). The 1st derivative coefficient of the fluctuation function also indicated a more complex profile (Fig. 9*g*). Nonetheless, kernel density estimation confirmed a clear local minimum, accurately identifying the crossover scale at $\log_{10} \hat{\tilde{s}}_c = 1.33$ (Fig. 9*h*). The two peaks of the density distribution reflected the two

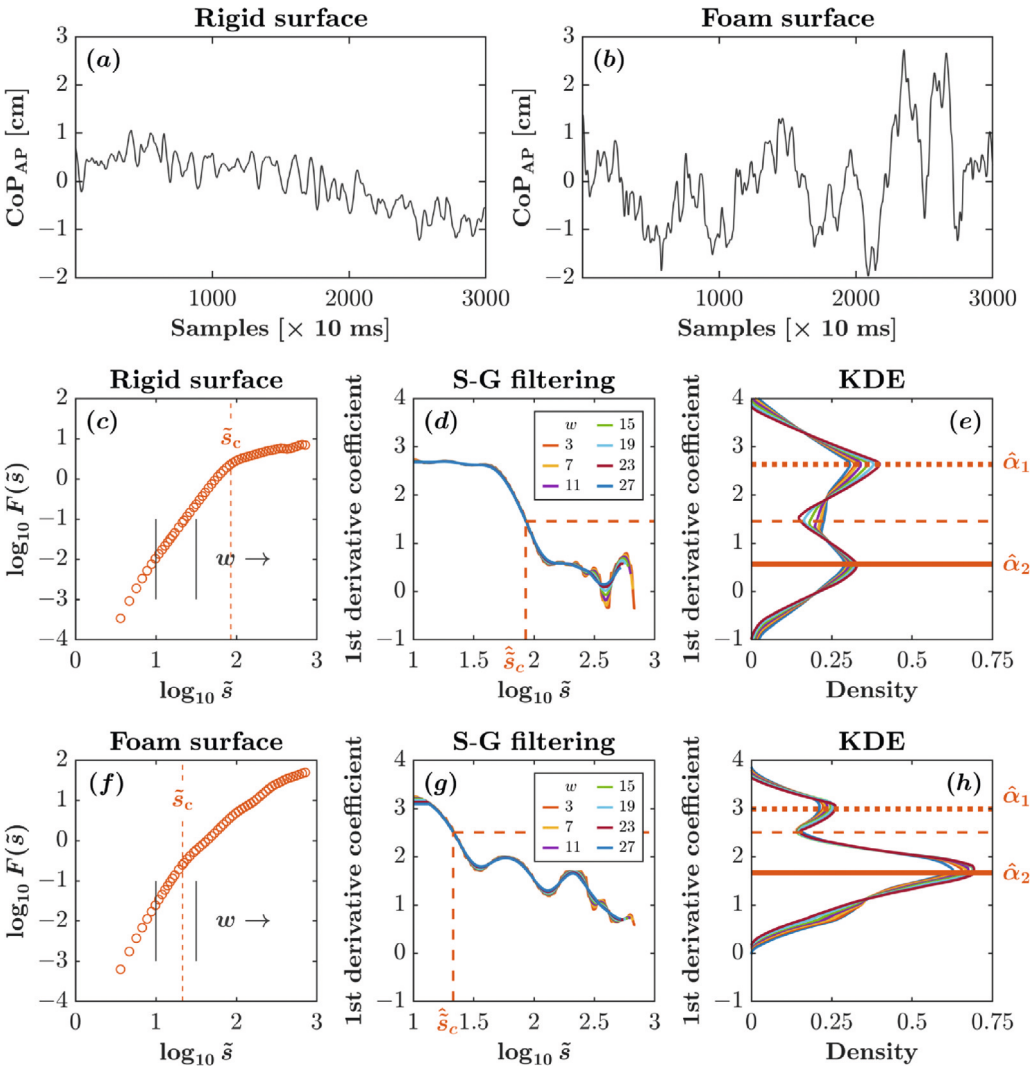


Fig. 9. Illustration of crossover estimation in postural CoP dynamics using the proposed combination of the Savitzky–Golay differentiation filter and kernel density estimation, revealing effects of task constraints. (*a*) CoP trajectory along the anteroposterior (AP) axis in an example healthy adult standing quietly on a rigid surface with eyes closed (length, $N = 3{,}000$ samples, recorded at 100 Hz). (*b*) CoP trajectory along the AP axis in the same healthy adult standing quietly on a foam surface with eyes closed ($N = 3{,}000$ samples, recorded at 100 Hz). (*c*) Fluctuation function of the CoP trajectory in (*a*) obtained using DMA2. (*d*) 1st derivative coefficient of the S–G filtering process obtained while changing the window width $w = 3, 7, 11, \ldots, \lfloor N_t/8 \rfloor$, where N_t is the number of points in the fluctuation function. (*e*) Density distribution of the 1st derivative coefficient of the S–G filter estimates the crossover scale, $\hat{\tilde{s}}$c, as the scale, $\tilde{s}$, corresponding to the 1st derivative coefficient at the trough of the density distribution, and the two scaling exponents, $\hat{\alpha}_1$ and $\hat{\alpha}_2$, as the peaks of the density distribution. (*f*–*h*) Illustration of the same process for estimating the crossover scale and two scaling exponents is applied to the CoP trajectory in (*b*). Dashed vertical lines in (*c*, *d*, *f*, & *h*) indicate crossovers. Dotted horizontal lines in (*e* & *h*) indicate $\hat{\alpha}_1$. Solid horizontal lines in (*e* & *h*) indicate $\hat{\alpha}_2$.

scaling exponents, $\hat{\alpha}_1 = 3.25$ and $\hat{\alpha}_2 = 1.64$, encoding the speed of CoP at shorter and longer scales. This crossover at ~ 214 milliseconds indicates an earlier shift in the nervous system's postural control strategies. Balance is maintained at scales shorter than 214 milliseconds through rapid, reflexive adjustments, resulting in faster CoP displacements. Conversely, longer scales involve more long-term control processes, such as anticipatory and strategic adjustments, leading to slower CoP displacements. Standing on the foam surface recruits these shorter-scale processes due to increased demands and continual postural instability, necessitating anticipatory and strategic adjustments. Hence, the proposed method successfully identifies crossovers in simple and complex conditions.

While the previous example demonstrates the potential of the proposed method in discerning the effects of task constraints on scaling crossovers, empirical measurements of task constraints can sometimes be exaggerated and highly task-specific. To highlight the sensitivity of the proposed method to more subtle variations in physiological intermittency, we present another example focusing on the impact of neurodegenerative diseases on postural control. We analyzed the fluctuation functions of two CoP trajectories: one from a healthy older adult and one from an age-matched Parkinson's patient, standing on a rigid surface with eyes open [98,99]. Fig. 10, *a* & *b* shows the CoP trajectories for older adults and Parkinson's patients, respectively. The CoP trajectory for the older adult exhibited a clear crossover, $\log_{10} \tilde{s}$, that was visually perceptible (Fig. 10*c*). The 1st derivative coefficient of the fluctuation function also indicated a region with a drastic change in $\log_{10} F(\tilde{s})$ (Fig. 10*d*), and kernel density estimation confirmed a local minimum identifying the crossover scale at $\log_{10} \hat{\tilde{s}}_c = 1.98$, which translates to ~ 950 milliseconds, and the two peaks identifying the two scaling exponents, $\hat{\alpha}_1 = 2.39$ and $\hat{\alpha}_2 = 1.18$, encoding the speed of CoP (Fig. 10*e*).

In contrast, the CoP trajectory for the Parkinson's patient exhibited an almost flat fluctuation profile at longer scales following the crossover (Fig. 10*f*). The 1st derivative coefficient of the fluctuation function reflected this crossover as a steep decline (Fig. 10*g*). Kernel density estimation confirmed a deep local minimum, accurately identifying the crossover scale at $\log_{10} \hat{\tilde{s}}_c = 2.03$ (Fig. 10*h*). The two peaks of the density distribution reflected the two scaling exponents,

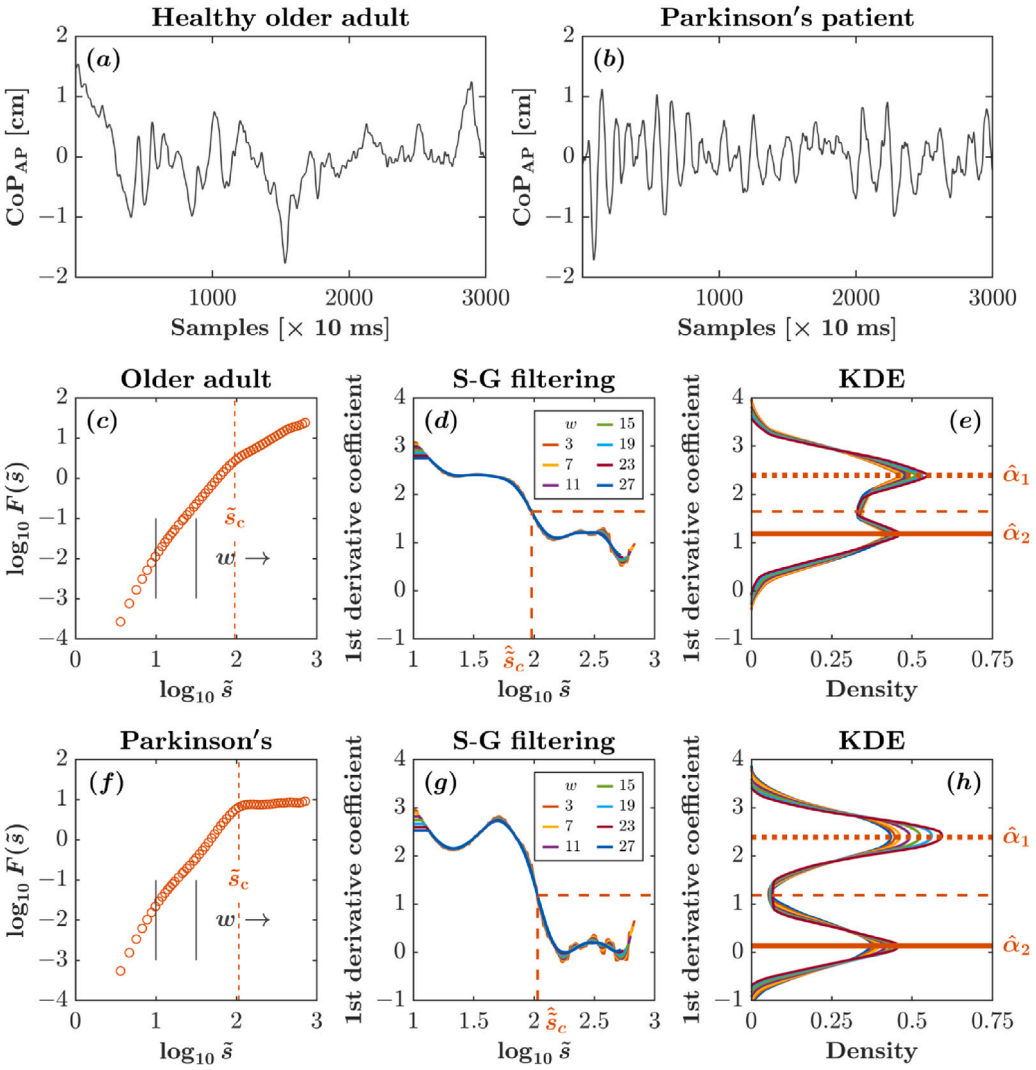


Fig. 10. Illustration of crossover estimation in postural CoP dynamics using the proposed combination of the Savitzky–Golay differentiation filter and kernel density estimation, revealing effects of neurodegenerative conditions. (*a*) CoP trajectory along the anteroposterior (AP) axis in an example healthy older adult standing quietly on a rigid surface with eyes open (length, $N = 3{,}000$ samples, recorded at 100 Hz). (*b*) CoP trajectory along the AP axis in an example age-matched Parkinson's patient standing quietly on a rigid surface with eyes open ($N = 3{,}000$ samples, recorded at 100 Hz). (*c*) Fluctuation function of the CoP trajectory in (*a*) obtained using DMA2. (*d*) 1st derivative coefficient of the S–G filtering process obtained while changing the window width $w = 3, 7, 11, \ldots, \lfloor N_f/8 \rfloor$, where N_f is the number of points in the fluctuation function. (*e*) Density distribution of the 1st derivative coefficient of the S–G filter estimates the crossover scale, $\hat{s}_c$, as the scale, $\tilde{s}$, corresponding to the 1st derivative coefficient at the trough of the density distribution, and the two scaling exponents, $\hat{\alpha}_1$ and $\hat{\alpha}_2$, as the peaks of the density distribution. (*f*–*h*) Illustration of the same process for estimating the crossover scale and two scaling exponents is applied to the CoP trajectory in (*b*). Dashed vertical lines in (*c*, *d*, *f*, & *h*) indicate crossovers. Dotted horizontal lines in (*e* & *h*) indicate $\hat{\alpha}_1$. Solid horizontal lines in (*e* & *h*) indicate $\hat{\alpha}_2$.

$\hat{\alpha}_1 = 2.39$ and $\hat{\alpha}_2 = 0.13$, indicating no change in the speed of CoP at longer scales. This suggests that while the Parkinson's patient had almost intact rapid, reflexive adjustments, resulting in faster CoP displacements, they had compromised long-term control processes, such as anticipatory and strategic adjustments, leading to slower, possibly passive CoP displacements. This indicates the general rigidity of the body characteristic of Parkinson's disease, including axial rigidity and poor trunk coordination, which impair stability during upright standing [100–103]. Hence, we establish the applicability of the proposed method to identify not only task-related changes in scaling properties of postural control dynamics but also changes due to diseases affecting physiology [104,105].

In summary, understanding crossovers in postural control and elucidating the mechanisms of postural regulation and its interactions with task constraints can offer invaluable insights for clinical assessments and interventions. The proposed method can improve monitoring changes in crossover points over time. This serves as a valuable metric for evaluating the progression of aging and Parkinson's disease and the effectiveness of various interventions. For instance, if a rehabilitation program reduces the delay in the crossover point and increases the scaling exponents, it would indicate an improvement in the patient's reflexive control and overall balance. Specifically for Parkinson's patients, addressing the significantly delayed crossover point through specialized rehabilitation programs can enhance reflexive control, reduce instability, and ultimately lower the risk of falls.

6. Discussion

The numerical experiments demonstrate the superior capability of the proposed method by reducing scaling distortion and consequently enhancing the accuracy of crossover estimation, particularly for higher-order DFA and DMA, which is necessary to account for polynomial trends in empirical time series. When applied to the fluctuation function, the Savitzky–Golay differentiation filter offers a more accurate assessment of the two scaling exponents than traditional segmented regression. This improved performance is consistently observed across

various simulation processes, including AR(1), ARFIMA$(1,d,0)$, and ARFIMA$(2,d,0)$. As a result, the method enables a more objective assessment of both crossovers and distinct scaling exponents. Notably, the accuracy of crossover estimation significantly improves when crossovers occur at longer scales, which is often the case with empirical time series, especially when measurements span extended periods. This study investigated crossover behavior using AR and ARFIMA processes, common econometric models. For readers in physics, future work could apply our detection method to models such as fractional Brownian motion, multiplicative cascading processes, and multifractal random walks (MRW). These models often exhibit complex scaling properties and highly relevant crossovers in physical systems. Applying our approach to these models would demonstrate its broader utility in accurately identifying and characterizing crossover behavior across diverse contexts and disciplinary interests.

Supplementing numerical experiments with actual physiological time series enhanced the validity and applicability of the proposed method. While theoretical models and simulated data allow the exploration of hypotheses and demonstration of the method's utility, they may not fully capture the complexity and variability of real-world physiological systems. By incorporating actual physiological time series of postural CoP, we tested the method against genuine data, ensuring its grounding in reality. We observed no discrepancies between theoretical predictions and observed behaviors, demonstrating the method's capability to detect crossovers in long-range correlated physiological time series. Additionally, the method revealed crossovers in CoP fluctuations that might be missed using previous methods, leading to more robust and clinically relevant insights into physiological processes.

Historically, the approach has been to quantify the linear scaling exponent from the log–log plot in both DFA and DMA, based on the idealized model of scale-invariant structure in fluctuations. However, this linearity is an approximation and may not reflect actual biological and natural systems. The quantification of long-range correlations was initially appealing because it could describe the complex textures observed in the real world without relying on perfect Euclidean objects. Assuming a simple linear relationship, long-range correlations contradict the philosophy of embracing the inherent complexity of these correlations. In our approach, the detection of the crossover point is integrated with scaling range identification by analyzing deviations in the scaling exponents across different regions of the time series. This ensures that the crossover point is not treated in isolation but as part of a broader context that captures transitions between distinct scaling behaviors. Future extensions of this work could further refine this relationship by incorporating adaptive algorithms to dynamically delineate scaling ranges and enhance the precision of crossover point detection. An even more comprehensive understanding must consider the entire fluctuation function without *a priori* expectations about its form. Future research should move away from focusing solely on crossovers and instead utilize the full profile of the scaling region.

7. Concluding remarks

This study presents a novel approach for estimating crossover points in long-range correlated time series using DFA with higher-order polynomials and DMA enhanced by the Savitzky–Golay differentiation filter. By addressing the limitations of conventional methods, which often suffer from scale distortions and fail to identify crossover points when the fluctuation function grows gradually and curvilinearly, our method demonstrates improved accuracy in estimating crossover points and scaling exponents. Numerical experiments with AR and ARFIMA models confirm the efficacy of our approach in capturing long-range correlations and crossover phenomena. Applying this method to the postural CoP time series of healthy young adults standing on unstable surfaces reveals its practical utility in distinguishing between different postural control mechanisms. Findings underscore the potential of the proposed method to enhance understanding of complex dynamics in domains such as physiological signals and financial markets by providing more reliable and detailed estimates of scaling behavior and crossovers.

CRediT authorship contribution statement

Yudai Fujimoto: Writing – review & editing, Writing – original draft, Visualization, Software, Formal analysis, Conceptualization. **Madhur Mangalam:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis. **Ken Kiyono:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Autoregressive (AR) processes

A.1. AR(1) process

Let us consider the AR(1) process $x[i]$ given by

$$(1 - aL)\, x[i] = \epsilon[i], \tag{A.1}$$

where $\epsilon[i]$ is additive white Gaussian noise (awGn) with a mean of zero and variance σ^2, L is the lag operator such that $(L \cdot x[i] = x[i-1])$. Using the linear frequency response relationship, the power spectrum of this process is defined by

$$\begin{aligned} S_x(f) &= \frac{\sigma^2}{|1 - a\exp(i2\pi f)|^2} \\ &= \frac{\sigma^2}{a^2 + 1 - 2a\cos(2\pi f)}. \end{aligned} \tag{A.2}$$

Eq. (A.2) can be approximated as

$$S_x(f) = \begin{cases} \dfrac{\sigma^2}{(1-a)^2} & |f| < \dfrac{|a-1|}{2\pi\sqrt{a}} \\ \dfrac{\sigma^2}{4a}\,|\pi f|^{-2} & \text{otherwise,} \end{cases} \tag{A.3}$$

In this process, the crossover frequency f_c is given by

$$f_c = \frac{|a-1|}{2\pi\sqrt{a}}. \tag{A.4}$$

Fig. 6, *a* & *b* shows the sample time series and power spectrum of the AR(1) process. The power spectrum is a Lorentzian spectrum with a flat white-noise plateau for $|f| \to 0$ and a $1/f^2$ behavior in the high-frequency range.

A.2. ARFIMA(1,d,0) process

Let us consider the ARFIMA$(1,\mathrm{d},0)$ process $x[i]$ given by

$$(1 - aL)x[i] = (1 - L)^{-d}\epsilon[i], \tag{A.5}$$

where $\epsilon[i]$ is awGn with mean zero and variance σ^2, L is the lag operator such that $L \cdot x[i] = x[i-1]$. In this process, the essential part of generating long-range correlated noise is the fractional difference operator $(1-L)^{-d}$, $-0.5 < d < 0.5$. Using the linear frequency response relationship, the power spectrum of this process is defined by

$$S_x(f) = \frac{\sigma^2\,|1 - \exp(i2\pi f)|^{-2d}}{|1 - a\,\exp(i2\pi f)|^2} \frac{\sigma^2\,|\sin(\pi f)|^{-2d}}{4^d\left[a^2 + 1 - 2a\cos(2\pi f)\right]}. \tag{A.6}$$

Eq. (A.6) can be approximated as

$$S_x(f) = \begin{cases} \dfrac{\sigma^2}{4^d\,(1-a)^2}\,|\pi f|^{-2d} & |f| < \dfrac{|a-1|}{2\pi\sqrt{a}} \\ \dfrac{\sigma^2}{4^{d+1}a}\,|\pi f|^{-2-2d} & \text{otherwise.} \end{cases} \tag{A.7}$$

In this process, the crossover frequency f_c is given by

$$f_c = \frac{a-1}{2\pi\sqrt{a}}. \tag{A.8}$$

When $d = 0$, this process is equivalent to an AR(1) process. Fig. 6, *c* & *d* shows the sample time series and power spectrum of the ARFIMA$(1,d,0)$ process. ARFIMA$(1,d,0)$ processes can also exhibit crossovers.

A.3. ARFIMA(2,d,0) process

Let us consider the ARFIMA$(2,\mathrm{d},0)$ process $x[i]$ given by

$$(1 - a_1L - a_2L^2)x[i] = (1-L)^{-d}\epsilon[i], \tag{A.9}$$

where $\epsilon[i]$ is awGn with a mean of zero and variance σ^2, L is the lag operator such that $L\cdot x[i] = x[i-1]$, and under the stationary condition, $-1 \le a_2 \le (1-|a_1|)$. Using the linear frequency response relationship, the power spectrum of this process is defined by

$$S_x(f) = \frac{\sigma^2\,|\sin(\pi f)|^{-2d}}{4^d\left[1 + a_1^2 + a_2^2 - 2a_1(1-a_2)\cos(2\pi f) - 2a_2\cos(4\pi f)\right]}. \tag{A.10}$$

In addition, when $a_1^2+4a_2 < 0$, the power spectrum has a mid-frequency peak at the frequency given by

$$f_p = \frac{1}{2\pi}\cos^{-1}\left\{\frac{a_1(a_2-1)}{4a_2}\right\}. \tag{A.11}$$

When we neglect this peak, Eq. (A.10) can be approximated as

$$S_x(f) = \begin{cases} \dfrac{\sigma^2}{4^d(1 + a_1^2 + a_2^2 - 2a_1(1-a_2) - 2a_2)}\,|\pi f|^{-2d} & |f| < f_c \\ \dfrac{3\sigma^2}{4^d\{4a_1(a_2-1) - 64a_2\}}\,|\pi f|^{-4-2d} & \text{otherwise,} \end{cases} \tag{A.12}$$

where the crossover frequency f_c is given by

$$f_c = \frac{3^{1/4}\left(1 - 2a_1 + a_1^2 - 2a_2 + 2a_1a_2 + a_2^2\right)^{1/4}}{\sqrt{2}\pi\left(a_1a_2 - a_1 - 16a_2\right)^{1/4}}. \tag{A.13}$$

Fig. 6, *e* & *f* shows the sample time series and power spectrum of the ARFIMA$(2,d,0)$ process.

Data availability

No data was used for the research described in the article.

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